A Periodic Table of the Finite Simple Groups

Every finite simple group appears exactly once, in its simplest form. Columns are grouped by rank; the vertical axis is the field size q.

TYPE

▭	Cyclic groups
▭	Alternating groups
▭	Chevalley groups
▭	Steinberg groups
▭	Suzuki and Ree groups
▭	Sporadic groups

HOW TO READ A CELL

$\cong A_2(2)$ — other descriptions of the same group

$A_1(7)$ — the group

$2^3 \cdot 3 \cdot 7$ — prime factorisation of its order

AN EMPTY CELL

the group is not defined there, or is not simple,
or it is isomorphic to one printed elsewhere —
every group appears only under its simplest name:

$A_1(1) \to C_1$
$A_1(4), A_1(5) \to A_5$ [$A_4(1)$]
$A_1(9) \to A_6$ [$A_5(1)$]
$A_2(2) \to A_1(7)$
$A_3(2) \to A_8$ [$A_7(1)$]
${}^2A_3(2) \to B_2(3)$
$C_n(q)$, q even $\to B_n(q)$
$B_2(2)' \to A_6$ [$A_5(1)$]
$G_2(2)' \to {}^2A_2(3)$
${}^2G_2(3)' \to A_1(8)$

FAMILIES

$A_n(q) = L_{n+1}(q)$ — Linear
$B_n(q) = O_{2n+1}(q)$ — Orthogonal (odd)
$C_n(q) = S_{2n}(q)$ — Symplectic
$D_n(q) = O^+_{2n}(q)$ — Orthogonal (plus)
${}^2A_n(q) = U_{n+1}(q)$ — Unitary
${}^2D_n(q) = O^-_{2n}(q)$ — Orthogonal (minus)
$G_2(q)$ — Exceptional G_2
$F_4(q)$ — Exceptional F_4
$E_6(q)$ — Exceptional E_6
$E_7(q)$ — Exceptional E_7
$E_8(q)$ — Exceptional E_8
${}^3D_4(q)$ — Triality
${}^2E_6(q)$ — Twisted E_6
${}^2B_2(q)$ — Suzuki
${}^2G_2(q)$ — Ree (small)
${}^2F_4(q)$ — Ree (large)

cyclic
rank 1
rank 2
rank 3
rank 4
rank 5
rank 6
rank 7
rank 8
rank 9
rank 10
rank n
1
2
3
4
5
7
8
9
11
13
16
17
19
23
25
27
29
31
32
243
q

q = size of the field

Sporadic groups

	Group	Order
Mathieu	M_{11}	$2^4 \cdot 3^2 \cdot 5 \cdot 11$
Mathieu	M_{12}	$2^6 \cdot 3^3 \cdot 5 \cdot 11$
Mathieu	M_{22}	$2^7 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11$
Mathieu	M_{23}	$2^7 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 23$
Mathieu	M_{24}	$2^{10} \cdot 3^3 \cdot 5 \cdot 7 \cdot 11 \cdot 23$
Janko	J_1	$2^3 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 19$
Hall–Janko	J_2	$2^7 \cdot 3^3 \cdot 5^2 \cdot 7$
Janko	J_3	$2^7 \cdot 3^5 \cdot 5 \cdot 17 \cdot 19$
Janko	J_4	$2^{21} \cdot 3^3 \cdot 5 \cdot 7 \cdot 11^3 \cdot 23 \cdot 29 \cdot 31 \cdot 37 \cdot 43$
Higman–Sims	HS	$2^9 \cdot 3^2 \cdot 5^3 \cdot 7 \cdot 11$
McLaughlin	McL	$2^7 \cdot 3^6 \cdot 5^3 \cdot 7 \cdot 11$
Held	He	$2^{10} \cdot 3^3 \cdot 5^2 \cdot 7^3 \cdot 17$
Rudvalis	Ru	$2^{14} \cdot 3^3 \cdot 5^3 \cdot 7 \cdot 13 \cdot 29$
Suzuki	Suz	$2^{13} \cdot 3^7 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13$
O'Nan	O'N	$2^9 \cdot 3^4 \cdot 5 \cdot 7^3 \cdot 11 \cdot 19 \cdot 31$
Conway	Co_1	$2^{21} \cdot 3^9 \cdot 5^4 \cdot 7^2 \cdot 11 \cdot 13 \cdot 23$
Conway	Co_2	$2^{18} \cdot 3^6 \cdot 5^3 \cdot 7 \cdot 11 \cdot 23$
Conway	Co_3	$2^{10} \cdot 3^7 \cdot 5^3 \cdot 7 \cdot 11 \cdot 23$
Fischer	Fi_{22}	$2^{17} \cdot 3^9 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13$
Fischer	Fi_{23}	$2^{18} \cdot 3^{13} \cdot 5^2 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 23$
Fischer	Fi_{24}'	$2^{21} \cdot 3^{16} \cdot 5^2 \cdot 7^3 \cdot 11 \cdot 13 \cdot 17 \cdot 23 \cdot 29$
Harada–Norton	HN	$2^{14} \cdot 3^6 \cdot 5^6 \cdot 7 \cdot 11 \cdot 19$
Lyons	Ly	$2^8 \cdot 3^7 \cdot 5^6 \cdot 7 \cdot 11 \cdot 31 \cdot 37 \cdot 67$
Thompson	Th	$2^{15} \cdot 3^{10} \cdot 5^3 \cdot 7^2 \cdot 13 \cdot 19 \cdot 31$
Baby Monster	B	$2^{41} \cdot 3^{13} \cdot 5^6 \cdot 7^2 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 31 \cdot 47$
Monster	M	$2^{46} \cdot 3^{20} \cdot 5^9 \cdot 7^6 \cdot 11^2 \cdot 13^3 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 \cdot 41 \cdot 47 \cdot 59 \cdot 71$

CLASSIFICATION THEOREM

Every finite simple group is cyclic of prime order, alternating of degree n ≥ 5, a group of Lie type, or one of the 26 sporadic groups. This poster shows every one of them, each exactly once.

NOTES

- After the cyclic column the columns are grouped by rank: every family with a simple group of rank 1, then of rank 2, and so on.
- E_8 is the last family to appear, so from rank 9 onwards every block is the same six classical families — hence the ellipsis.
- Row q = 1: over “the field with one element” a group of Lie type degenerates to its Weyl group, and the cell $A_n(1)$ holds the alternating group A_{n+1}. Only type A is simple there, so that row is exactly the alternating groups.
- A cell carrying a parameter shows the generic order formula in place of a prime factorisation, and so does a cell whose factorisation is too long to print.
- The rows past q = 13 are selected larger prime powers, chosen so that the Suzuki and Ree columns — which exist only for $q = 2^{2m+1}$ and $q = 3^{2m+1}$ — carry more than one group each.
- $C_n(q) \cong B_n(q)$ for q even, so only the odd-q cells of the C columns are printed.
- ${}^2B_2(q)$ and ${}^2F_4(q)$ exist for $q = 2^{2m+1}$ and ${}^2G_2(q)$ for $q = 3^{2m+1}$ (m ≥ 1); their other rows are empty.
- ${}^2F_4(2)$ is not simple, but its derived subgroup — the Tits group — is, and is printed in that cell.
- The 26 sporadic groups belong to no family of Lie type; they are shown in their own strip.
- A symbol with a subscript and no field is a group over “F_1”: C_p is cyclic of order p and A_m is alternating of degree m. With a field, $A_n(q)$ is linear and $C_n(q)$ symplectic — the argument tells them apart.